EasySeva

Government Services, Made Easier

A Citizen First Web Portal. A Human Computer Interaction Design Report

A Project Report submitted for the course

Human Computer Interaction

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Live application: `https://easyseva-rho.vercel.app`

Source repository: `https://github.com/rZk1809/easyseva`

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1 Abstract

EasySeva is a responsive web portal that helps ordinary citizens find, apply for, and track government services in a Tamil Nadu setting. People can use it in their own language and move at their own pace. We built it as a Human Computer Interaction project, and we designed it mainly for first time users and for people who are not very comfortable with technology. These are exactly the users who often find official portals confusing, full of jargon, and easy to get wrong.

This report explains what the application does, why it is useful, and how we applied well known HCI principles on every screen. We look at the interface through four lenses that complement each other. These are Jakob Nielsen's ten usability heuristics, Ben Shneiderman's eight golden rules, Don Norman's principles of interaction design, and the WCAG accessibility pillars known as POUR. For each page of the application we show a screenshot, describe what the page is for, and list the principles we used. For every principle we say what it means and how it shows up on that screen. The report ends with a coverage table and a short reflection on how good design helps real people have a better experience.

2 Introduction

2.1 What is EasySeva

Applying for a government document is something almost every family has to do at some point. It could be an income certificate, a scholarship, a ration card, or an old age pension. Yet the online experience is usually hard. Forms often put thirty fields on a single page, use acronyms that are never explained, work only in English, give you no way to save your progress, and report status in words that only an official would understand. For a first time user, or for someone who is not confident with computers, all of this becomes a wall.

EasySeva rethinks that experience around the way citizens actually think, rather than the way departments are organised. It offers the following.

- A catalogue of nine services across six citizen friendly categories. These are Certificates, Benefits, Education, Family, Senior Citizen, and Document Updates.
- A guided flow that moves through an eligibility check, a document checklist, a short form broken into steps, a review, and a confirmation.
- Passwordless login using a one time code, the ability to save a draft and come back later, and tracking by reference number.
- A trilingual interface in English, Tamil, and Hindi, along with built in accessibility controls for larger text, higher contrast, and reduced motion.

2.2 Who is it for and why is it useful

The people we designed for are the ones who usually feel uneasy with bureaucracy. They include first time applicants, elderly users, and people with limited English or limited digital experience. EasySeva is useful because it lowers every barrier that normally causes a person to give up or make a mistake. It speaks plain language, it shows exactly what is needed before

the user starts, it checks the input as it is typed, it never throws away the user's work, and it explains every status in human terms. The result is that a citizen can complete an application correctly on the first try, and can do so with confidence and dignity.

2.3 How HCI enables a good interaction

Human Computer Interaction is about designing systems around what people can do, what they struggle with, and what they expect. A government portal is a very good example of a task where this matters a lot. The task is high stakes, it happens rarely, and it tends to make people anxious. The user may touch the system only once a year, cannot afford a mistake, and has had no training. When we apply ideas that are well understood in this field, such as keeping status visible, preventing errors, favouring recognition over memory, giving the user control, and designing for everyone, the whole experience changes. An anxious and error prone chore becomes calm, predictable, and forgiving. The rest of this report shows, screen by screen, how we did that.

3 HCI Principles and Frameworks Used

This section defines the frameworks we refer to in the page by page analysis.

3.1 Nielsen's 10 Usability Heuristics

N1. **Visibility of system status.** Keep users informed with timely and appropriate feedback.

N2. **Match between the system and the real world.** Speak the users' language with familiar words and ideas.

N3. **User control and freedom.** Offer clear exits, undo, and the freedom to change course.

N4. **Consistency and standards.** Make similar things look and behave in similar ways.

N5. **Error prevention.** Design so that problems do not happen in the first place.

N6. **Recognition rather than recall.** Keep options and information visible so people do not have to remember them.

N7. **Flexibility and efficiency of use.** Let both new and experienced users work in the way that suits them.

N8. **Aesthetic and minimalist design.** Show only what is relevant and remove clutter.

N9. **Help users recognise, diagnose, and recover from errors.** Explain errors in plain language and offer a way forward.

N10. **Help and documentation.** Provide help that is easy to find and focused on the task.

3.2 Shneiderman's 8 Golden Rules

S1. Strive for consistency.

S2. Cater to universal usability, from novices to experts, including accessibility.

S3. Offer informative feedback.

S4. Design dialogs to yield closure.

S5. Prevent errors.

S6. Permit easy reversal of actions.

S7. Keep users in control.

S8. Reduce short term memory load.

3.3 Norman's Interaction Design Principles

D1. **Affordances and signifiers.** Controls look like what they do and signal how to use them.

D2. **Feedback.** Every action produces a result the user can perceive.

D3. **Mapping.** The link between a control and its effect feels natural, for example a progress bar that matches the steps completed.

D4. **Constraints.** The design limits invalid actions.

D5. **Consistency.** Similar operations use similar elements.

D6. **Conceptual model.** The interface helps users build a correct picture of how the process works.

3.4 WCAG Accessibility Pillars, known as POUR

Content should be Perceivable, Operable, Understandable, and Robust. EasySeva addresses these through contrast aware theming, full keyboard operation with visible focus, plain language content, and semantic markup that assistive technology can read.

4 Design System and Foundations

Before the page analysis, three foundations are worth noting because they support consistency, universal usability, and accessibility on every screen.

4.0.1 A consistent visual language

Every screen is built from a small set of reusable pieces, namely Button, Card, Input, Alert, Dialog, Badge, and Progress. They all draw on shared design tokens, including a fixed colour palette, consistent spacing and corner radius, and a single typeface. Because of this, a primary action, a status badge, or a form field looks and behaves the same way everywhere. Learning one screen effectively teaches the user every screen.

4.0.2 Built in accessibility controls

Each user can adapt the interface to suit them. There are three text sizes, a high contrast theme, and a reduced motion mode that respects the operating system preference. A skip to main content link, visible keyboard focus rings, large touch targets, and ARIA roles and labels are present throughout.

4.0.3 A trilingual interface

The entire interface is available in English, Tamil, and Hindi. The language can be chosen on the landing page or switched at any time from the header, and the choice is remembered on the next visit.

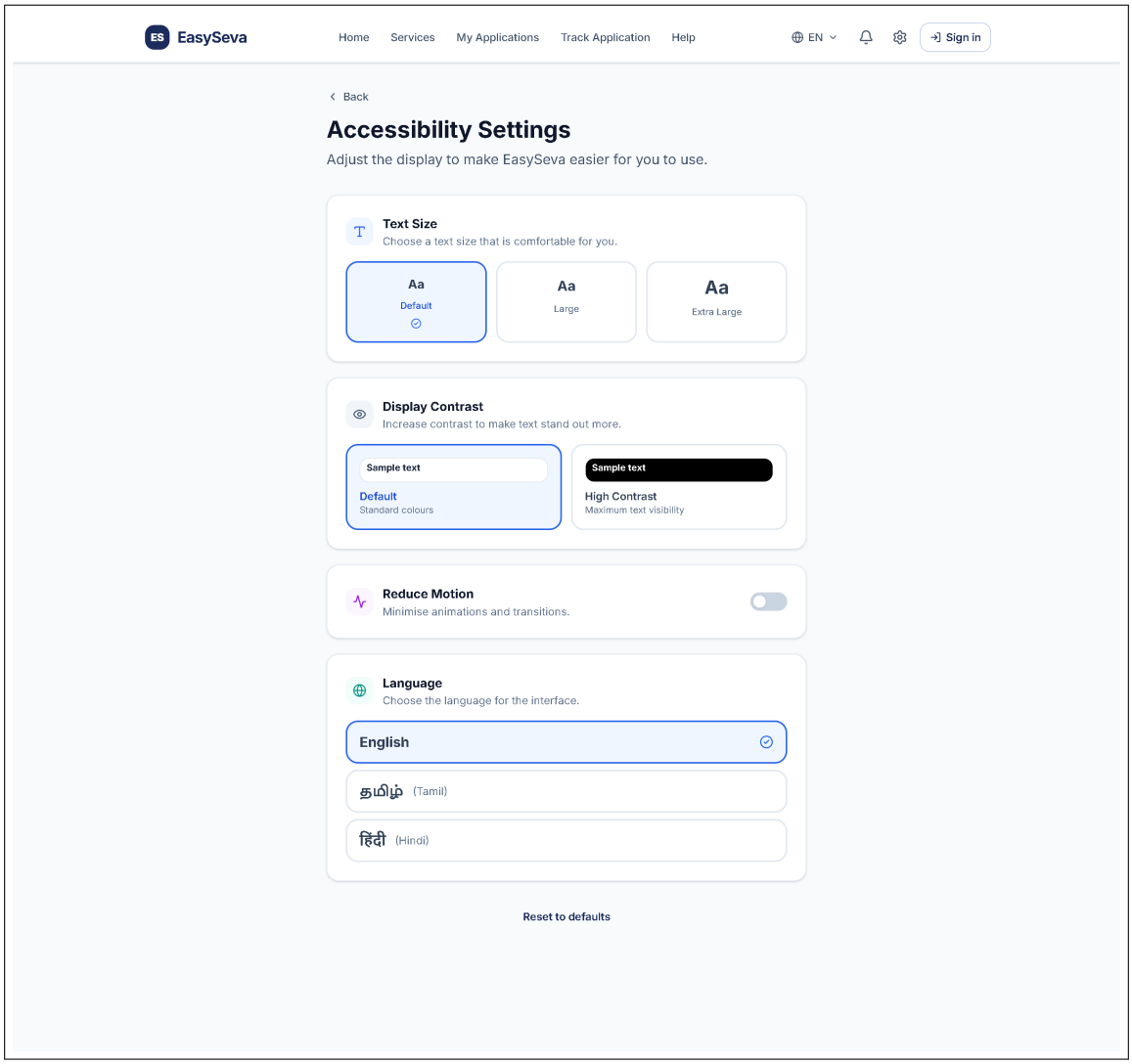


Figure 1. The Accessibility Settings page lets each user adjust text size, contrast, motion, and language. This is a direct example of Shneiderman's rule to cater to universal usability.

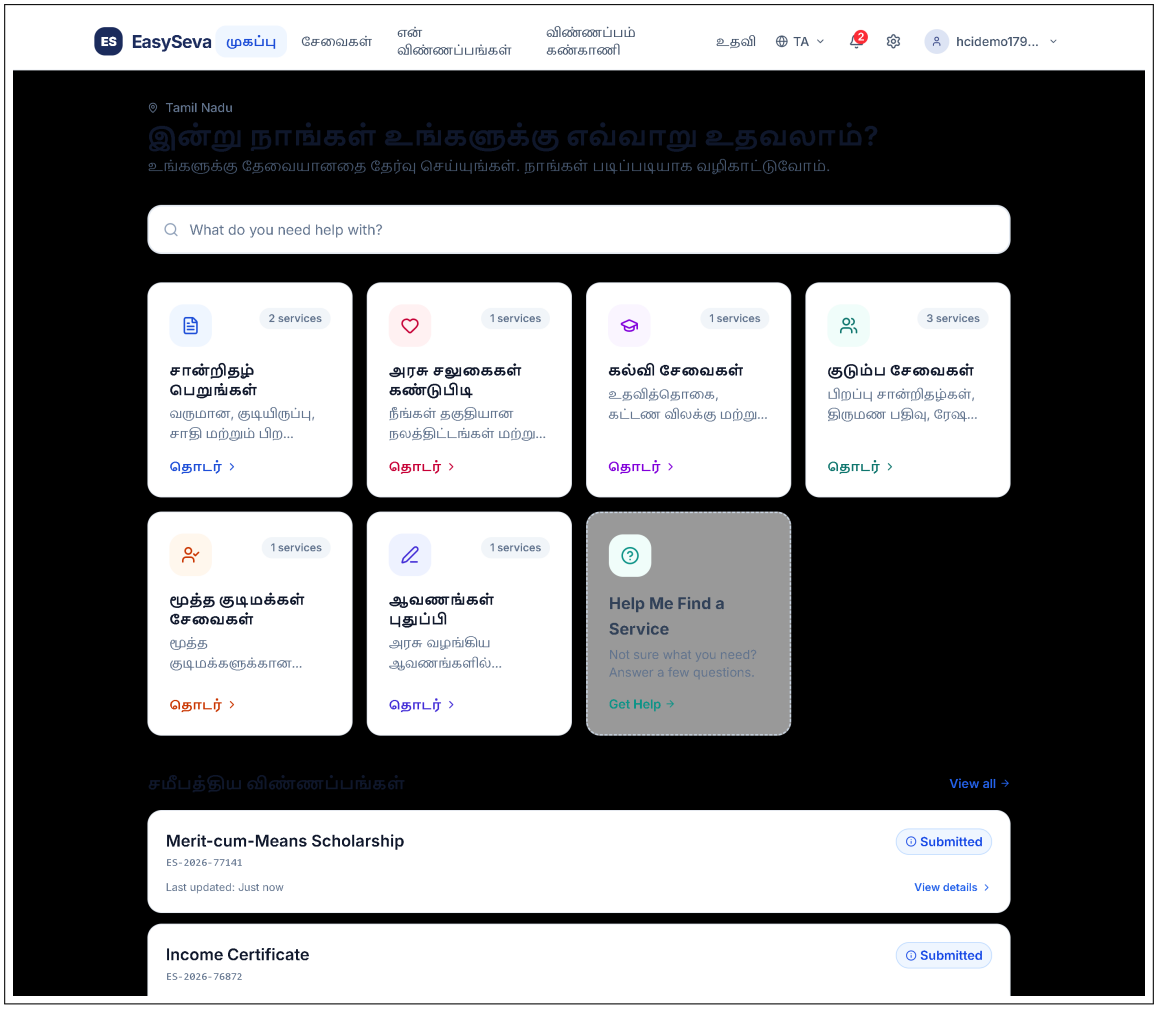


Figure 2. High contrast mode, shown here together with the Tamil interface. The dark high contrast theme improves readability for users with low vision, and it also shows that the accessibility and language settings apply across the whole app and stay in place as the user moves between pages.

5 Page by Page HCI Analysis

For each screen we state what it is for and then list the principles we applied. For every principle we give its meaning and the concrete way it appears on the screen.

5.1 Landing Page

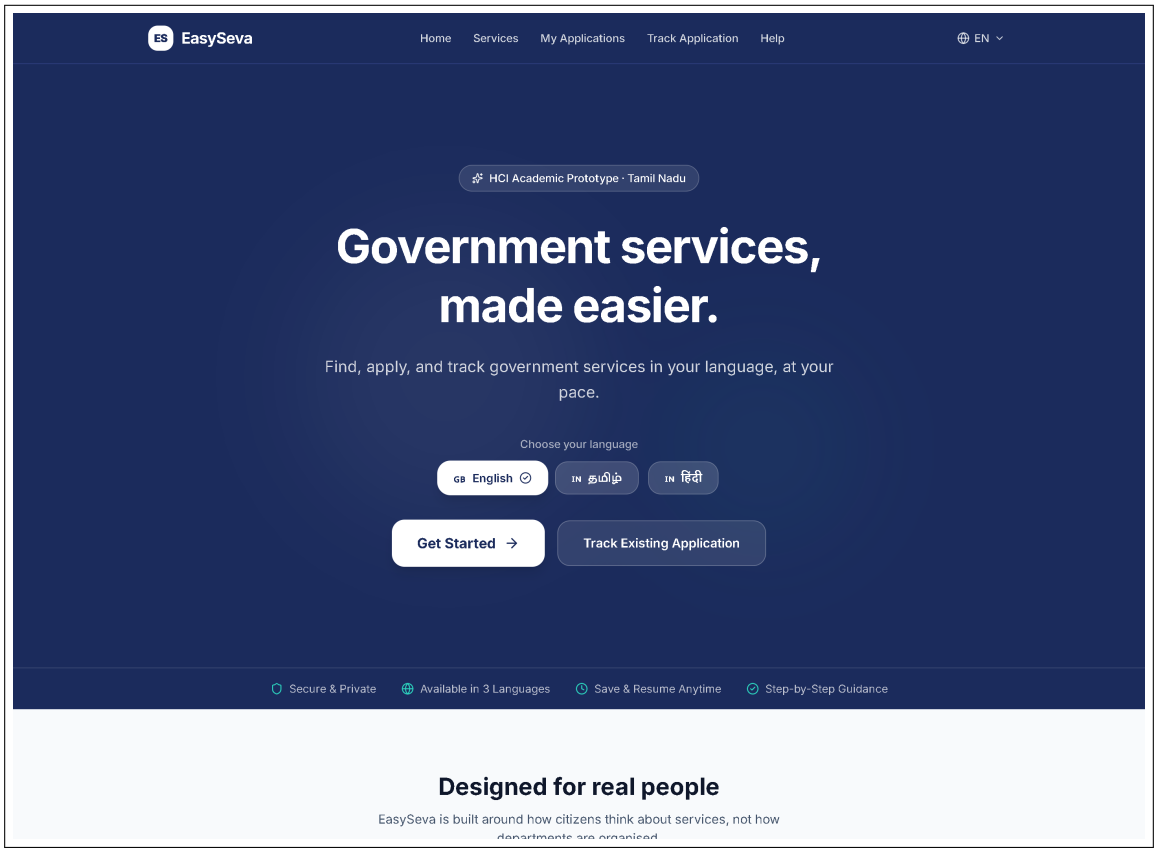


Figure 3. The landing page offers one clear message, an up front language choice, and two obvious next actions.

Purpose. Earn trust in a few seconds and send the user to the one thing they came to do, which is to start a service or track an existing one.

- **Aesthetic and minimalist design (N8).** *Show only what matters.*
  There is a single headline, one supporting line, the language choice, and exactly two buttons. Nothing competes for the first time user's attention.
- **Match with the real world (N2).** *Use the user's language.*
  The promise that government services are made easier, and that you can find, apply, and track them in your language and at your pace, is written the way a citizen thinks rather than in departmental terms.
- **Flexibility and efficiency (N7).** *Serve different users.*
  The language is chosen before anything else, so a Tamil or Hindi speaker is never pushed through an English wall, as Figure 4 shows.
- **Recognition rather than recall (N6).** *Make the options visible.*
  The trust strip, which lists secure and private, three languages, save and resume, and step by step guidance, surfaces the key capabilities instead of expecting the user to discover them.

- **Affordance and signifiers (D1).** *Controls should look usable.*
  The solid Get Started button clearly invites the main action, while the lighter Track button is visually secondary, so importance matches appearance.

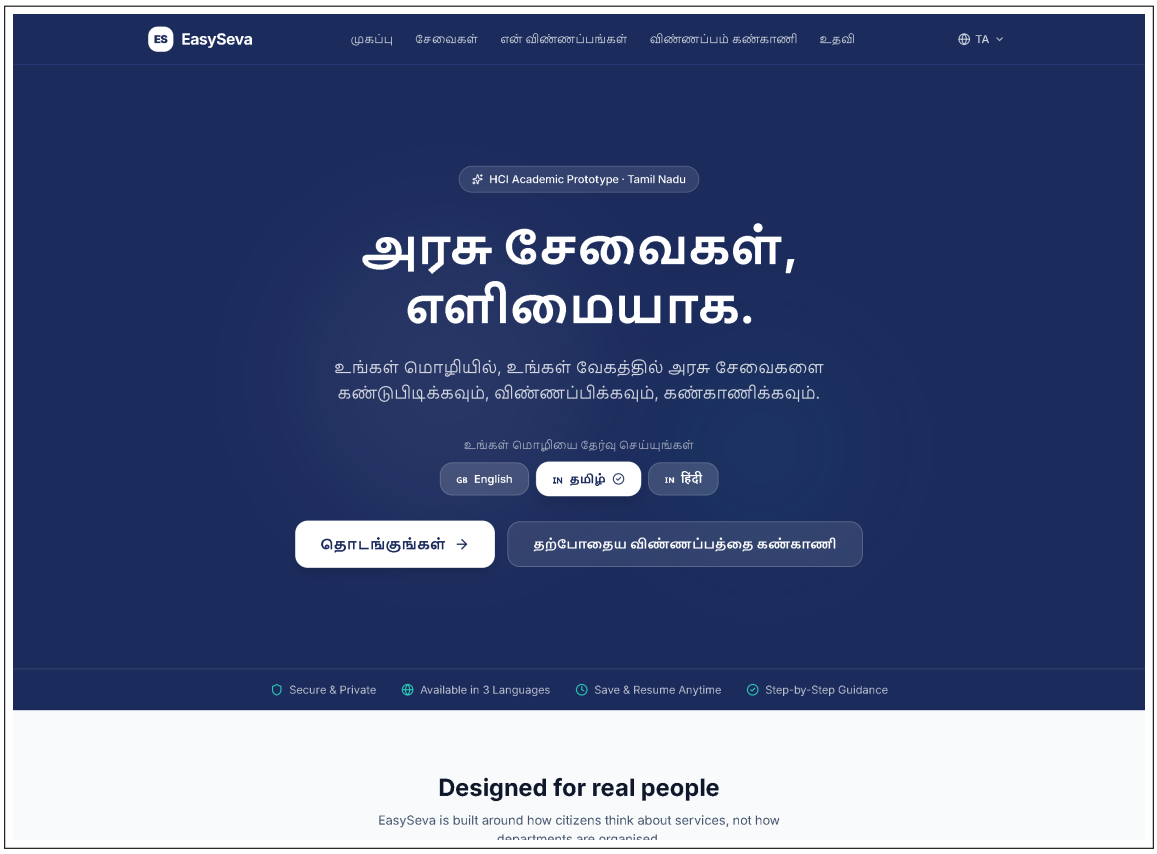


Figure 4. The same landing experience in Tamil. Choosing a language rewrites the whole interface, which directly supports match with the real world and flexibility of use for people who do not read English comfortably.

5.2 Home Dashboard

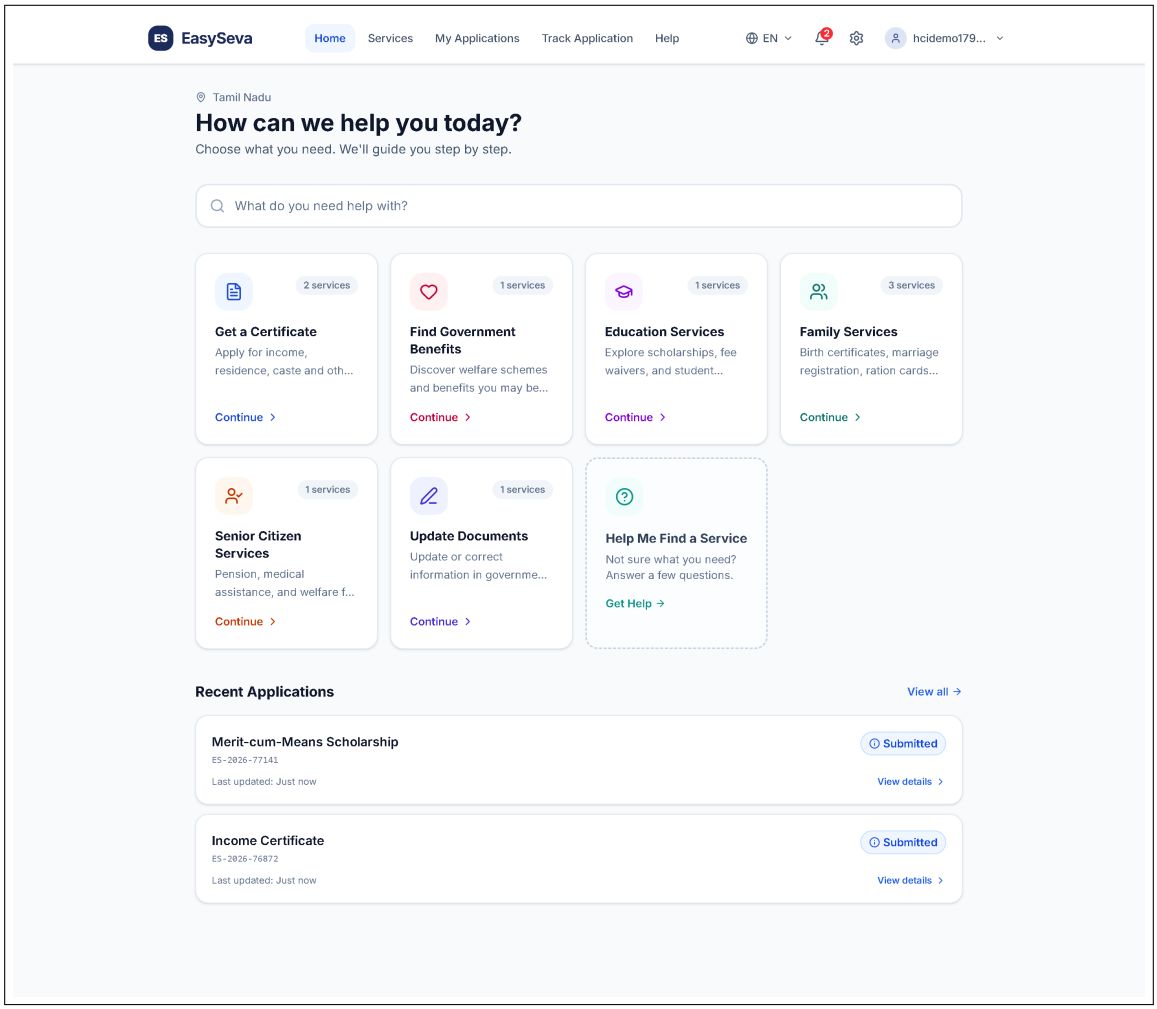


Figure 5. The home dashboard shows task based service categories, a natural language search box, a help me find a service option, and the user's recent applications with their current status.

Purpose. Give the signed in citizen one place to begin a new task or return to an existing one.

- **Match with the real world (N2).** *Organise around user goals.*
  Services are grouped as get a certificate, find government benefits, and family services, which are citizen intentions rather than department names.
- **Recognition rather than recall (N6).** *Show, do not ask people to remember.*
  Each category is a large card with an icon, a title, and example services, so the user recognises the right path instead of recalling a menu.
- **Visibility of system status (N1, S3).** *Keep users informed.*
  The recent applications section shows each reference number and its current status, such as submitted, so the user always knows where things stand.
- **Flexibility and efficiency (N7).** *Offer more than one route.*
  A confident user can type a need into the search box, while an unsure user can click help me find a service and be guided by a few questions.

5.3 Service Catalogue and Category Pages

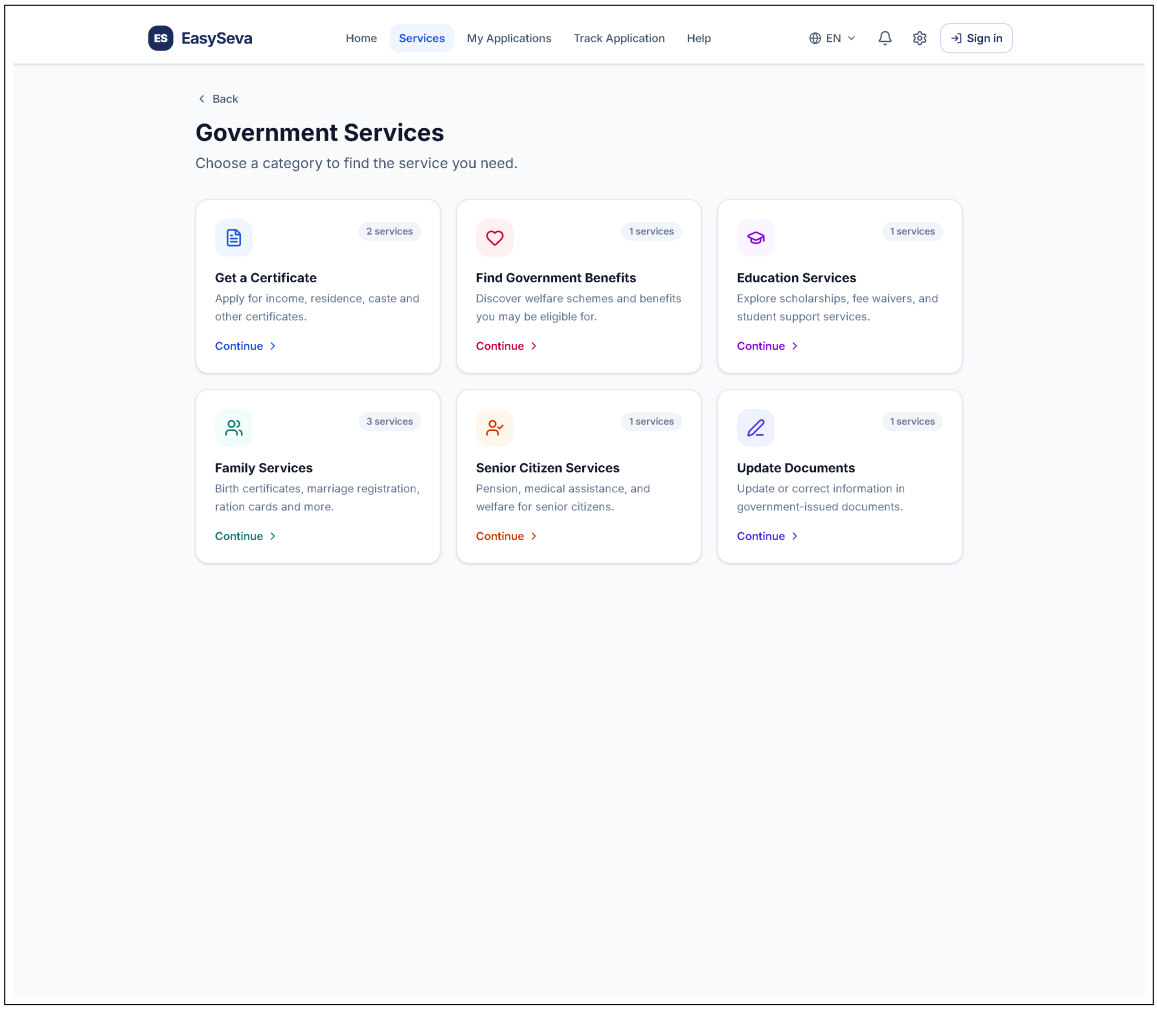


Figure 6. The full catalogue, grouped into the six citizen friendly categories, with an accurate count of the services in each one.

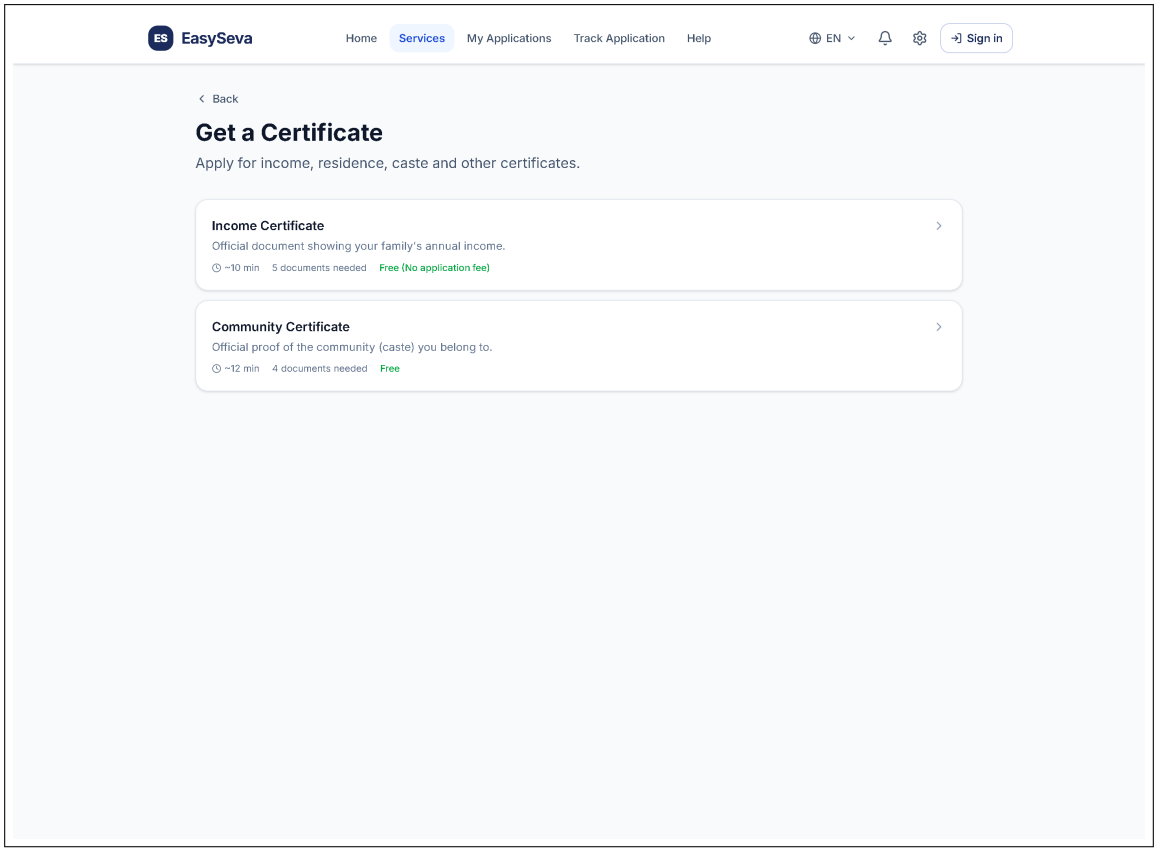


Figure 7. A category page, Certificates, listing the services it holds.

Purpose. Let users browse the whole catalogue or drill into one category.

- **Consistency and standards (N4, S1).** *Similar things look similar.*
  Every service and category uses the same card, so browsing feels predictable across the catalogue.
- **Recognition rather than recall (N6).** *Make the structure visible.*
  Category icons and short descriptions help users recognise where a service belongs.
- **Aesthetic and minimalist design (N8).** *Reduce overload.*
  A large catalogue is split into six easy groups rather than one long list.

5.4 Service Detail Page

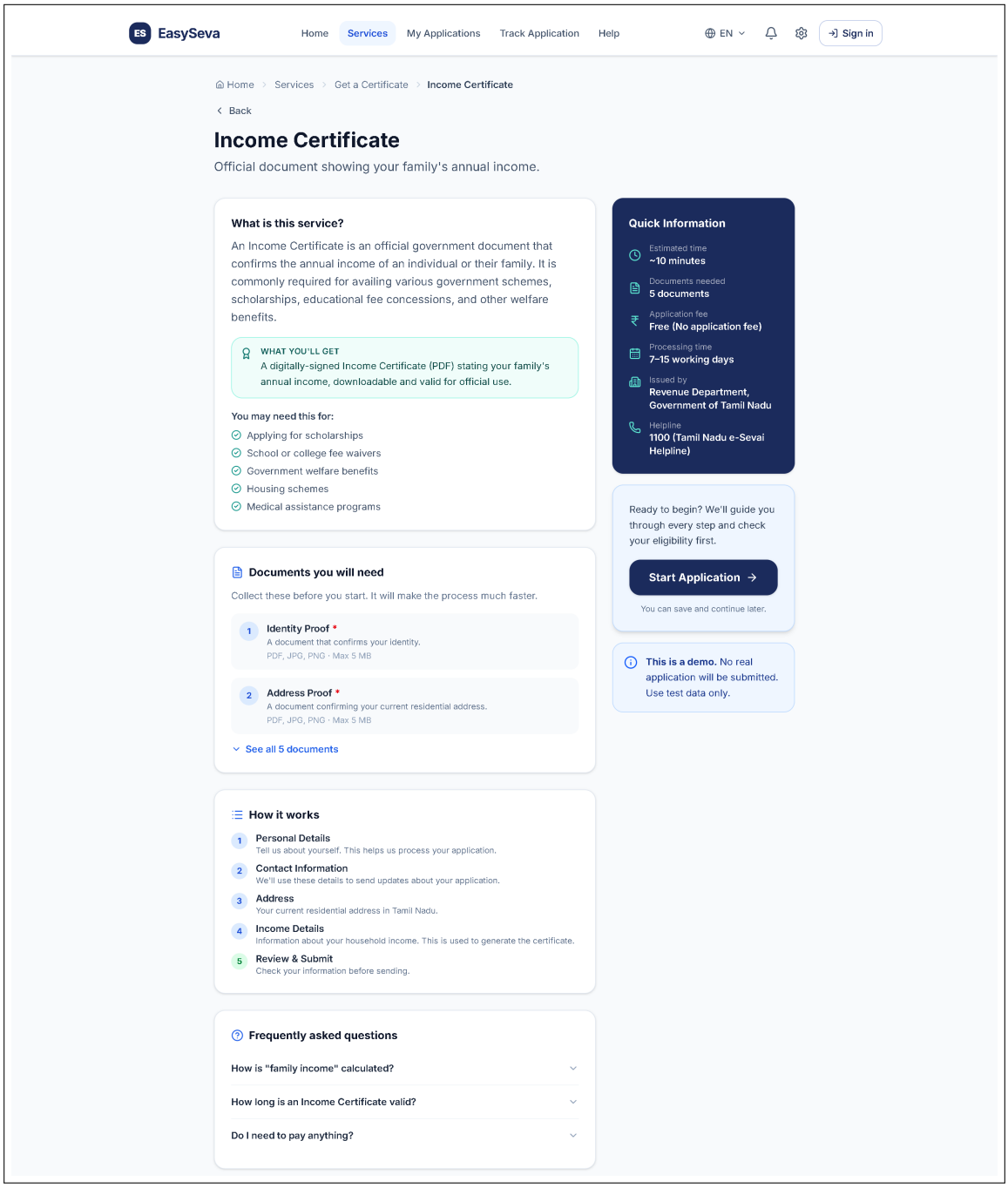


Figure 8. The service detail page for the Income Certificate. It shows breadcrumbs, a plain explanation, a what you will get note, a document checklist, a how it works overview, a set of questions and answers, and a quick information panel with time, fee, processing duration, the issuing department, and a helpline.

Purpose. Help the user decide, with confidence, whether and how to apply before they spend any time.

- **Help and documentation (N10).** *Put help where it is needed.*
  The what is this service note, the you may need this for list, and a dedicated questions and answers section answer the citizen's real doubts at the moment of decision.

- **Visibility of system status (N1).** *Set expectations.*
  The quick information panel states the estimated time, the fee, the processing duration, the issuing department, and a helpline, so there are no surprises later.
- **Recognition rather than recall (N6).** *Keep the user oriented.*
  Breadcrumbs that read home, services, certificates, income certificate always show where the user is and let them return to any level in one click.
- **Error prevention (N5, S5).** *Let people prepare first.*
  The documents you will need checklist lets users gather everything before they begin, which prevents them from abandoning the form halfway through.
- **User control and freedom (N3, S7).** *Avoid forcing a commitment.*
  A gentle note that you can save and continue later, together with one clear primary action to start the application, keeps the user in charge.

5.5 Eligibility Check

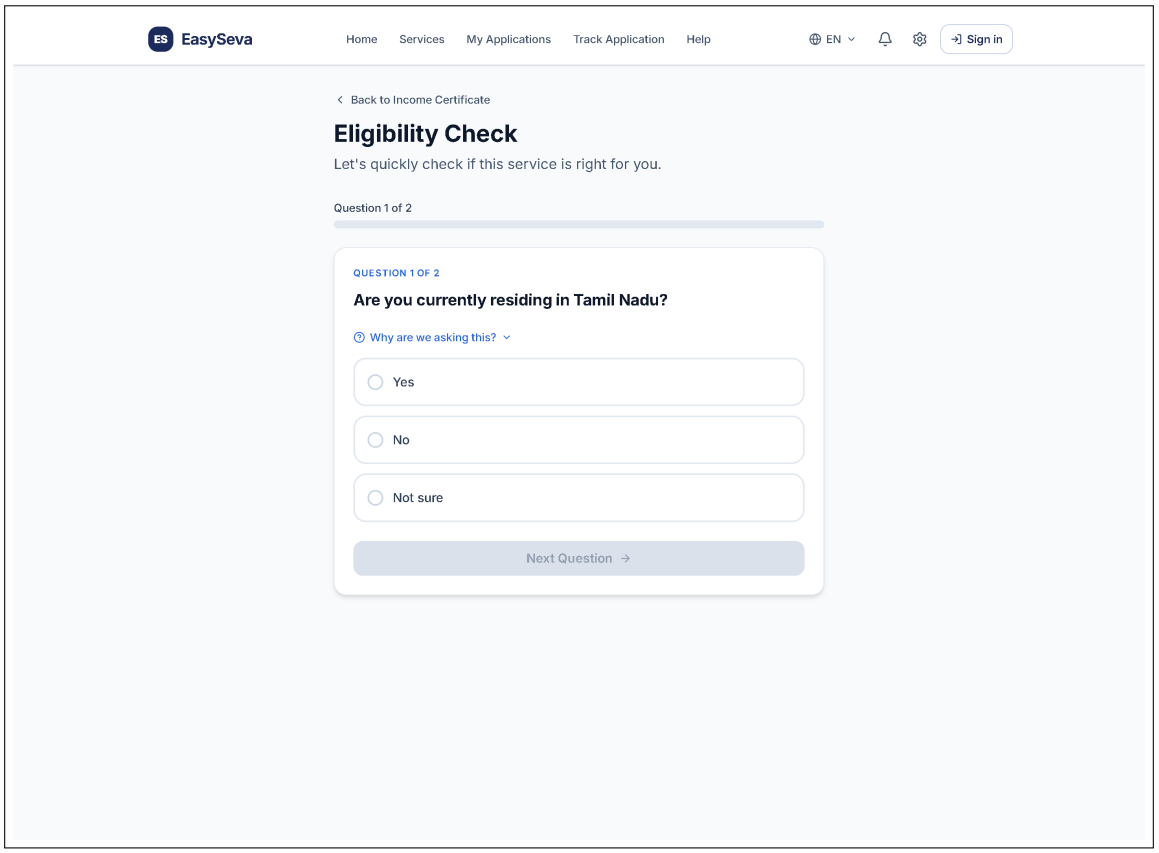


Figure 9. The eligibility check asks one plain question at a time, with an optional explanation of why we are asking, and a progress bar.

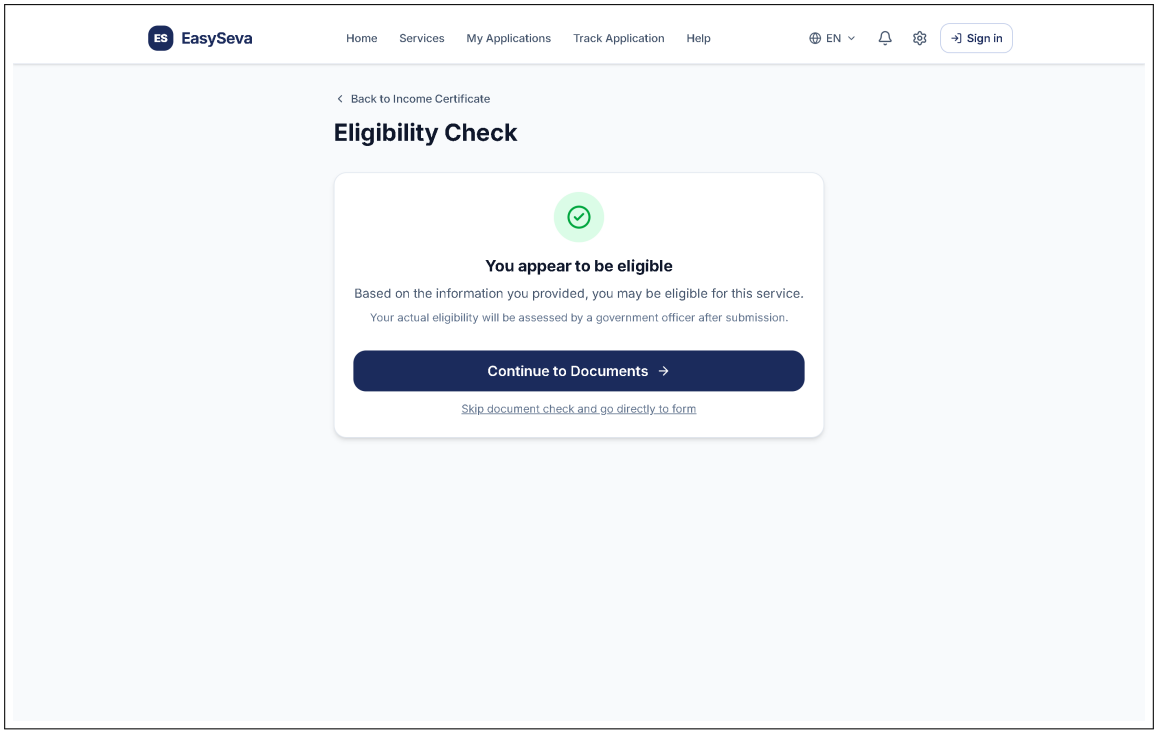


Figure 10. The result screen gives a clear verdict. Even when the user may not qualify, it preserves their freedom to continue or to look for a more suitable service.

Purpose. Save wasted effort by checking suitability early, without discouraging the user.

- **Error prevention (N5, S5).** *Stop problems early.*
  Checking eligibility before the form prevents the larger failure of filling in everything and then being rejected.
- **Help and documentation (N10).** *Explain the question.*
  Each question offers a why are we asking this note, which builds understanding and trust.
- **Visibility of system status (N1, D3).** *Map progress clearly.*
  A question one of two progress bar maps directly to how far the user has come.
- **User control and freedom (N3, S7).** *Respect the user's choice.*
  An ineligible result never blocks the user. They can continue anyway and let an officer decide, or ask for help finding a different service.
- **Informative feedback (S3).** *Announce the outcome.*
  The result sits in a live region, so a screen reader announces the verdict right away.

5.6 Document Upload

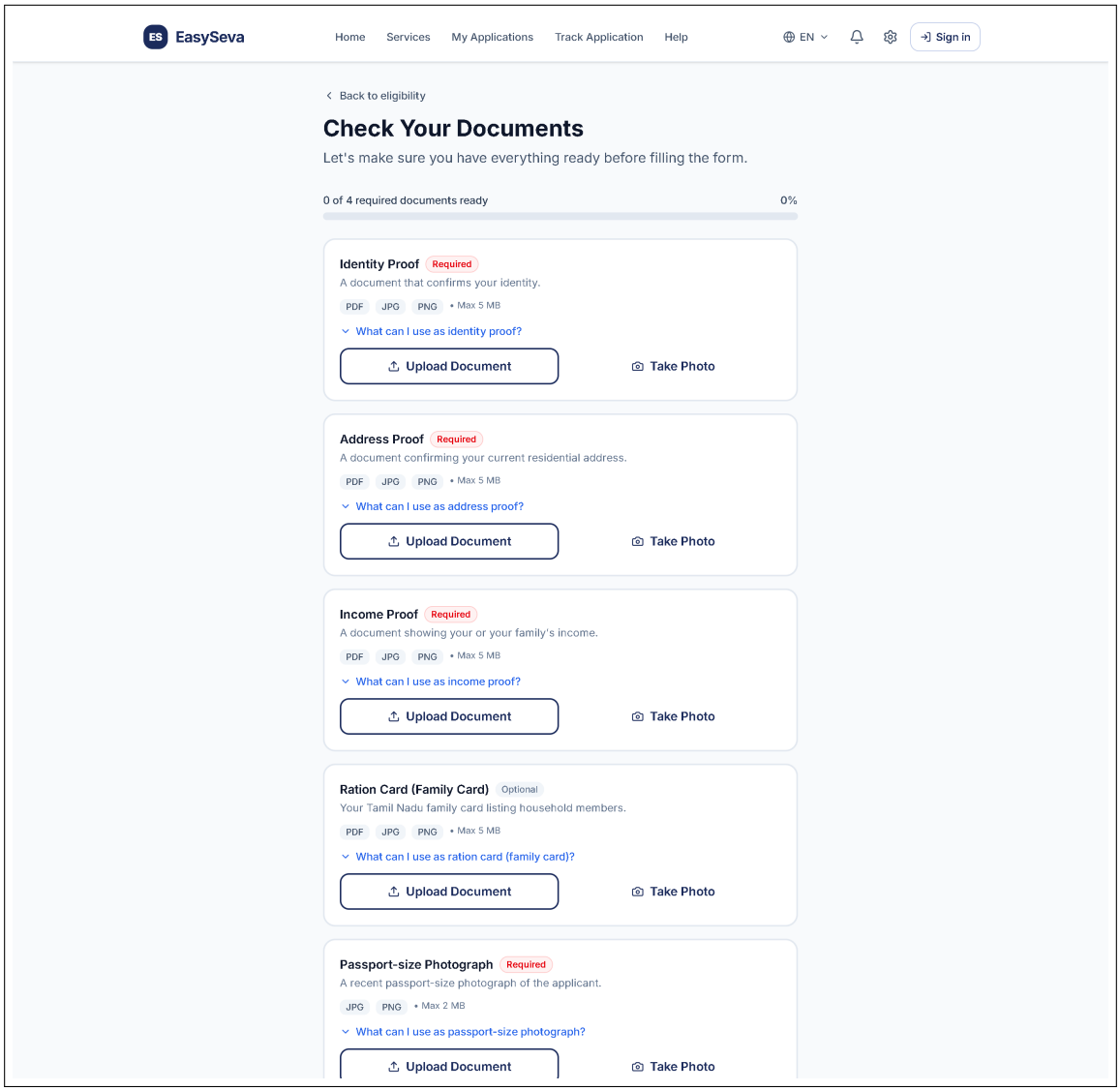


Figure 11. The document step. Each requirement is its own card with the accepted formats, a size limit, examples of acceptable documents, and upload or take photo actions.

Purpose. Make uploading the right documents easy and free of errors.

- **Recognition rather than recall (N6).** *Show the acceptable options.*
  A what can I use note lists concrete examples such as Aadhaar or Voter ID, so the user does not have to remember what qualifies.
- **Error prevention (N5, S5).** *Check before accepting.*
  Only PDF, JPG, or PNG files within the size limit are accepted, and invalid files are rejected with a clear reason before anything is submitted.
- **Visibility of system status (N1, S3).** *Confirm success.*
  An uploaded document turns green with a tick and shows its file name and size, so the user can see it was accepted.

- **Flexibility and efficiency (N7).** *Fit the device.*
  Both an upload option and a take photo option are offered, and the second one is ideal on a phone.
- **Constraints (D4).** *Guide valid input.*
  The file picker is limited to the accepted file types, which nudges the user toward a valid choice.

5.7 Application Form

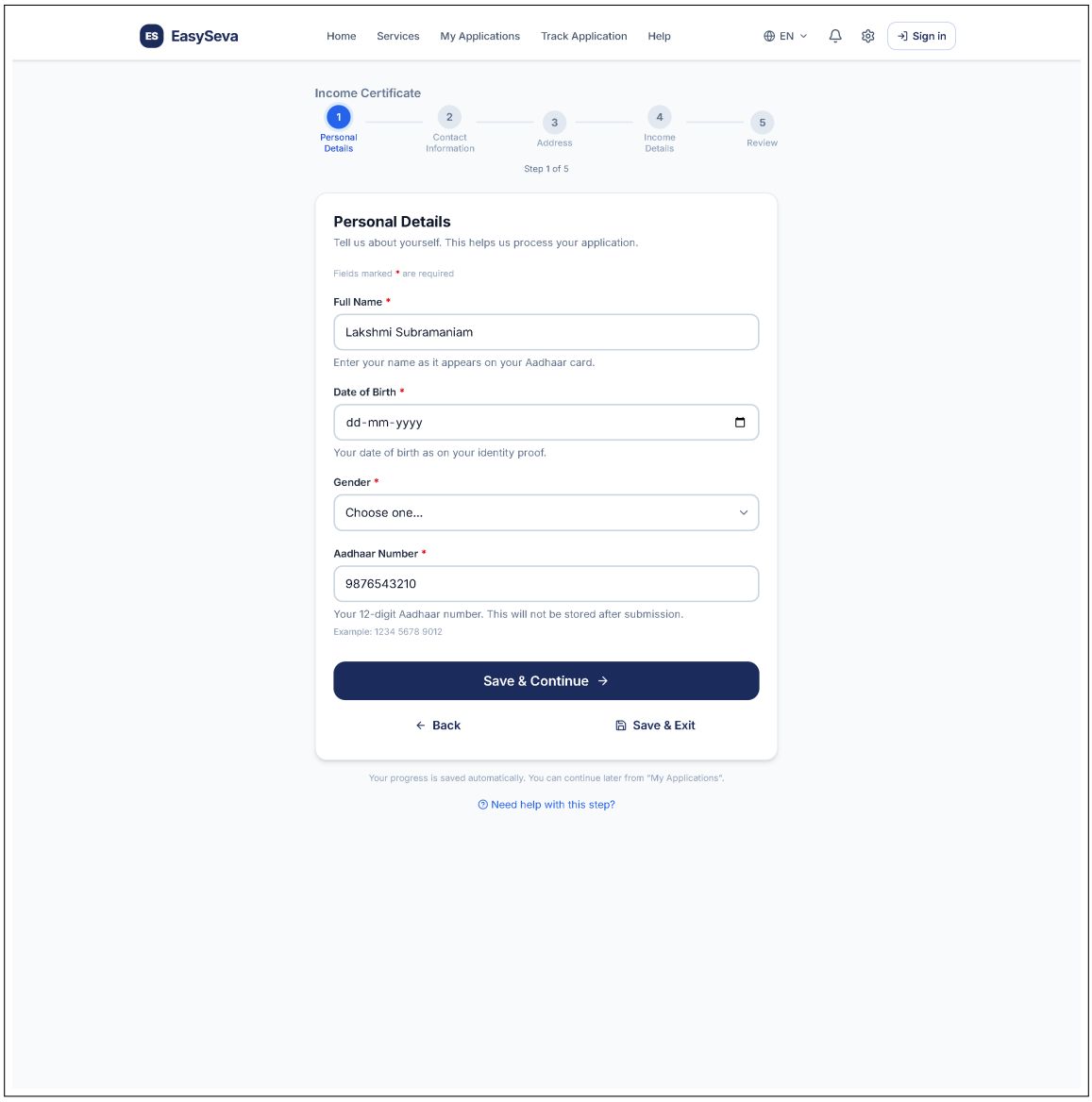


Figure 12. The form presents one small step at a time, with plain labels, helper text, examples, and inline checking. It never shows a wall of thirty fields.

Purpose. Collect accurate information with the least possible mental effort.

- **Reduce short term memory load (S8, N6).** *Break the work into pieces.*

The form is split into small themed steps for personal details, contact, address, and income, instead of one overwhelming page.

- **Error prevention (N5, S5).** *Catch mistakes as they happen.*
  Fields are checked as the user types, for example a twelve digit Aadhaar, a ten digit mobile number, or a six digit PIN code, each with a specific and friendly message.
- **Match with the real world (N2).** *Use human labels.*
  Labels and helper text use everyday phrasing, such as estimated annual income of the household, with worked examples.
- **Visibility of system status (N1, D3).** *Show progress.*
  A step indicator shows how many steps are left, which maps effort to completion.
- **User control and freedom (N3, S6).** *Never lose work.*
  A save and finish later option stores a draft at any point, so users can leave and return safely.

5.8 Review Before Submit

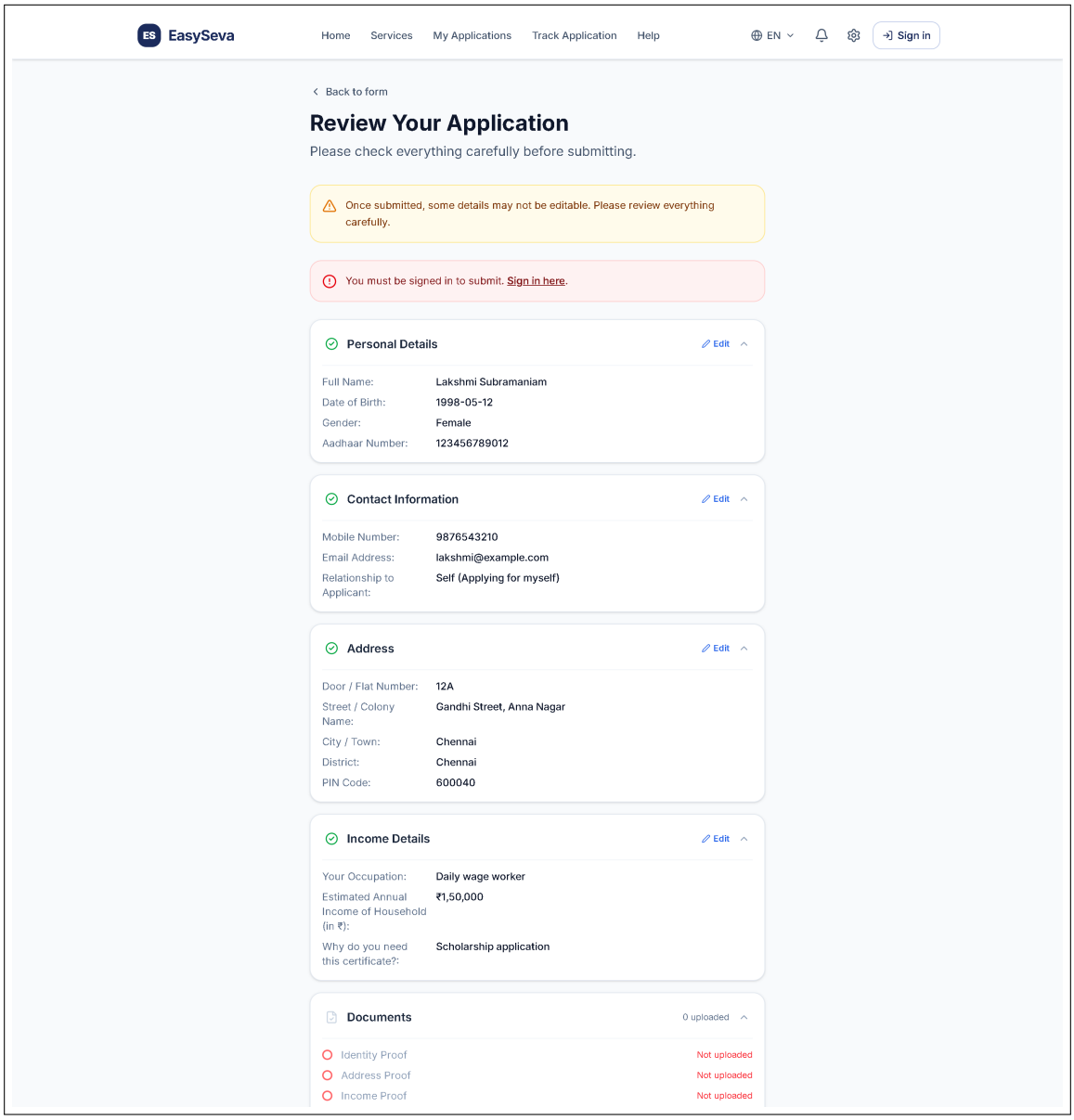


Figure 13. The review screen summarises every value the user entered and every document they uploaded, in collapsible sections. Each section can be edited on its own, and there is a final confirmation step before submission.

Purpose. Let the user check everything and fix anything before the submit, which cannot be undone.

- **Recognition rather than recall (N6).** *Summarise instead of quizzing.*
  Every answer and document is shown back to the user, so they verify by recognising their entries rather than trying to remember them.
- **User control and freedom (N3, S6).** *Make reversal easy.*
  An edit control on each section jumps straight back to that part of the form.
- **Error prevention (N5, S5).** *Confirm before committing.*

A warning banner and a confirmation dialog, which states that the action cannot be undone, guard the final submit.

- **Design dialogs to yield closure (S4).** *Mark a clear end.*
  The review gives a definite check then submit close to the data entry part of the task.

5.9 Confirmation

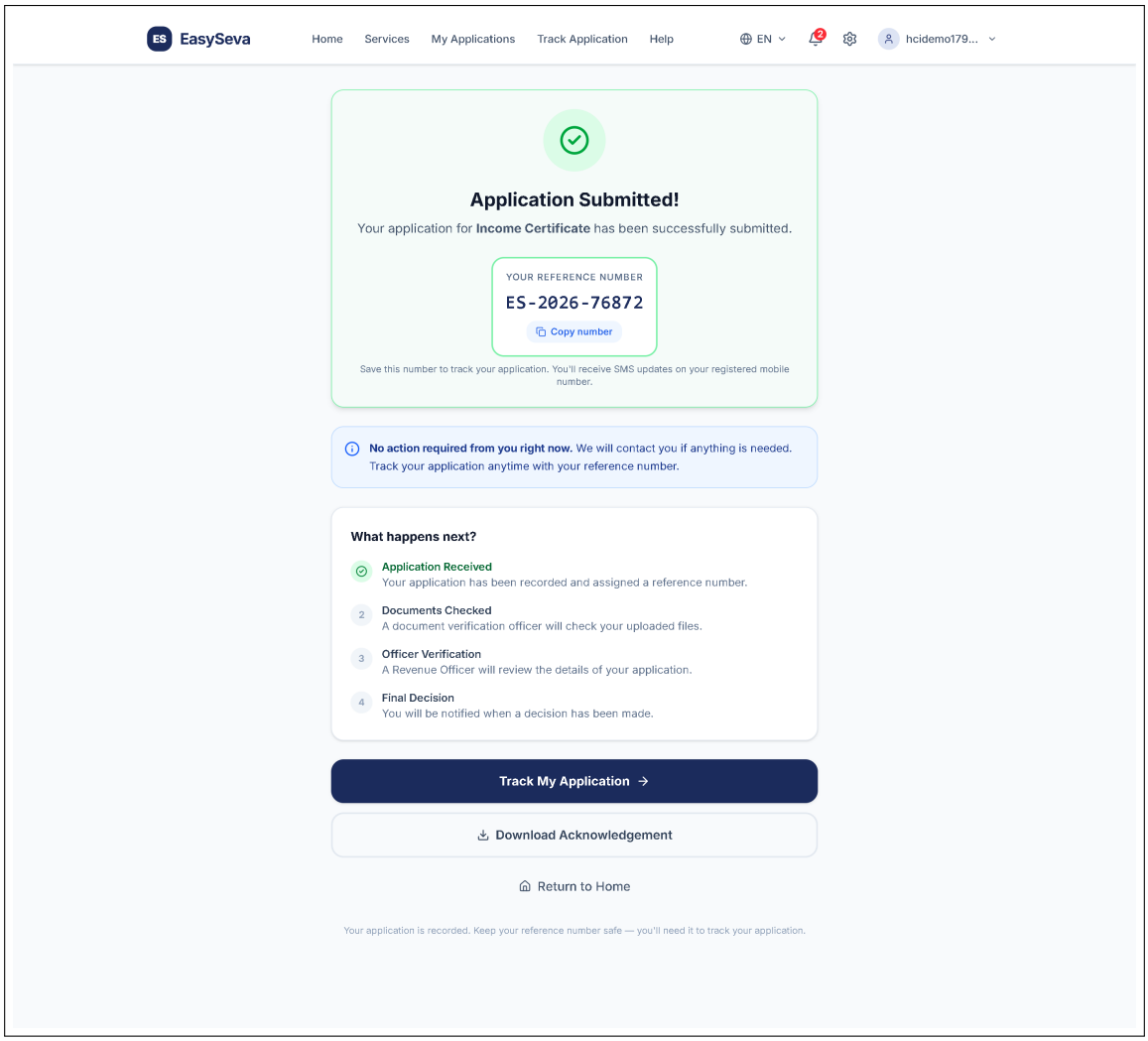


Figure 14. The confirmation screen acknowledges success, shows the reference number to save, and explains what happens next.

Purpose. Reassure the user that the application was received and tell them the next steps.

- **Visibility of system status (N1, S3).** *Make success clear.*
  A success state and a prominent reference number confirm that the submission went through.
- **Design dialogs to yield closure (S4).** *Complete the task.*
  The flow ends with a clear done state and a note on what happens next, which gives the user a sense of closure.

- **Help users plan the next step (N10).** *Tell them what follows.*
  Guidance on tracking and the likely timeline removes the common what now worry.

5.10 Track an Application

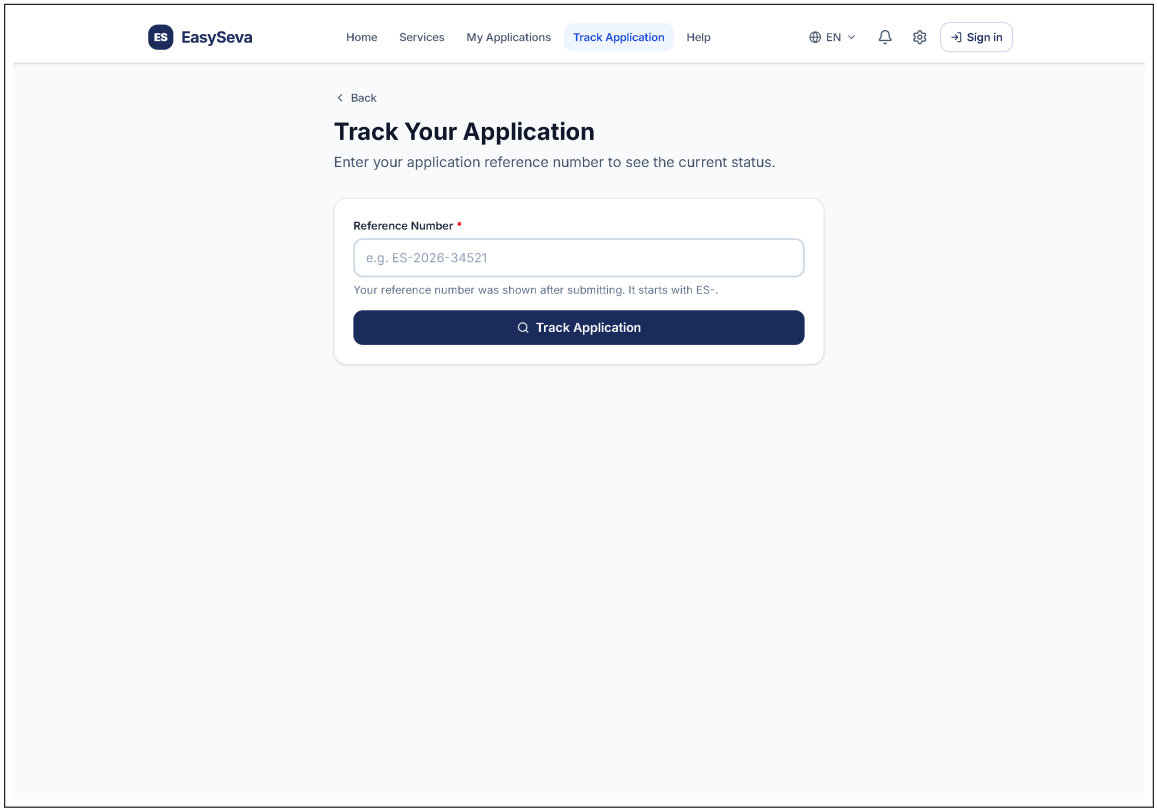


Figure 15. Tracking by reference number, which works even without signing in, returns a plain language status and a timeline.

Purpose. Let anyone check progress quickly, including people who applied on behalf of someone else.

- **Flexibility and efficiency (N7).** *Keep access low friction.*
  Tracking works without login, so a family member can check the status with just the reference number.
- **Match with the real world (N2).** *Use plain status words.*
  Statuses read as officer verification or documents verified, not as internal codes.
- **Visibility of system status (N1).** *Show progress at a glance.*
  A staged timeline shows exactly where the application is in the pipeline.

5.11 My Applications

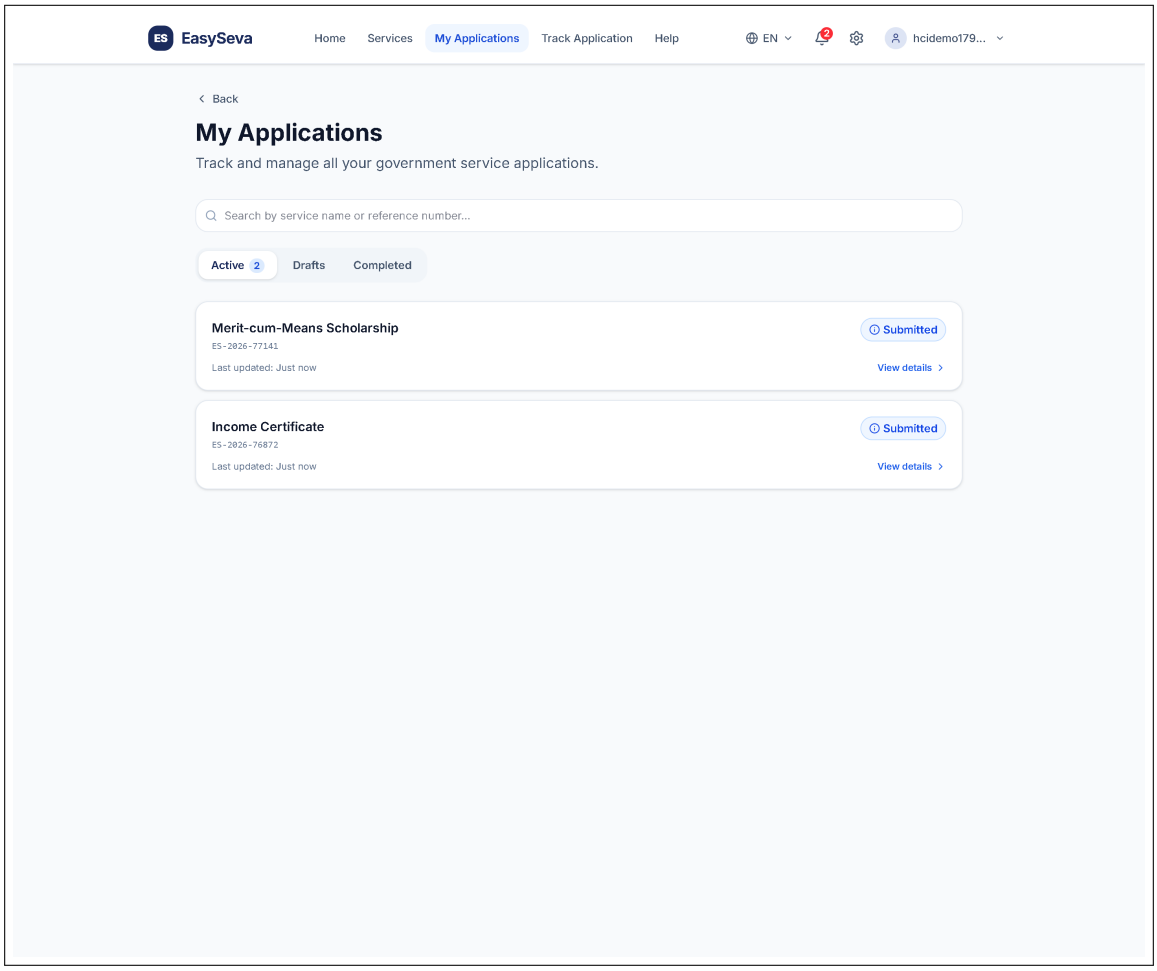


Figure 16. The applications list with active, drafts, and completed tabs, a search box, and a status badge on every item.

Purpose. A personal space to watch and resume all applications.

- **Visibility of system status (N1, S3).** *Put status everywhere.*
  Each item carries a colour coded status badge for an at a glance overview.
- **Flexibility and efficiency (N7).** *Filter and find.*
  Tabs separate active, drafts, and completed, and a search box finds items by name or reference number.
- **Consistency and standards (N4, S1).** *Reuse familiar patterns.*
  The same card and badge used elsewhere appear here, so there is nothing new to learn.

5.12 Notifications

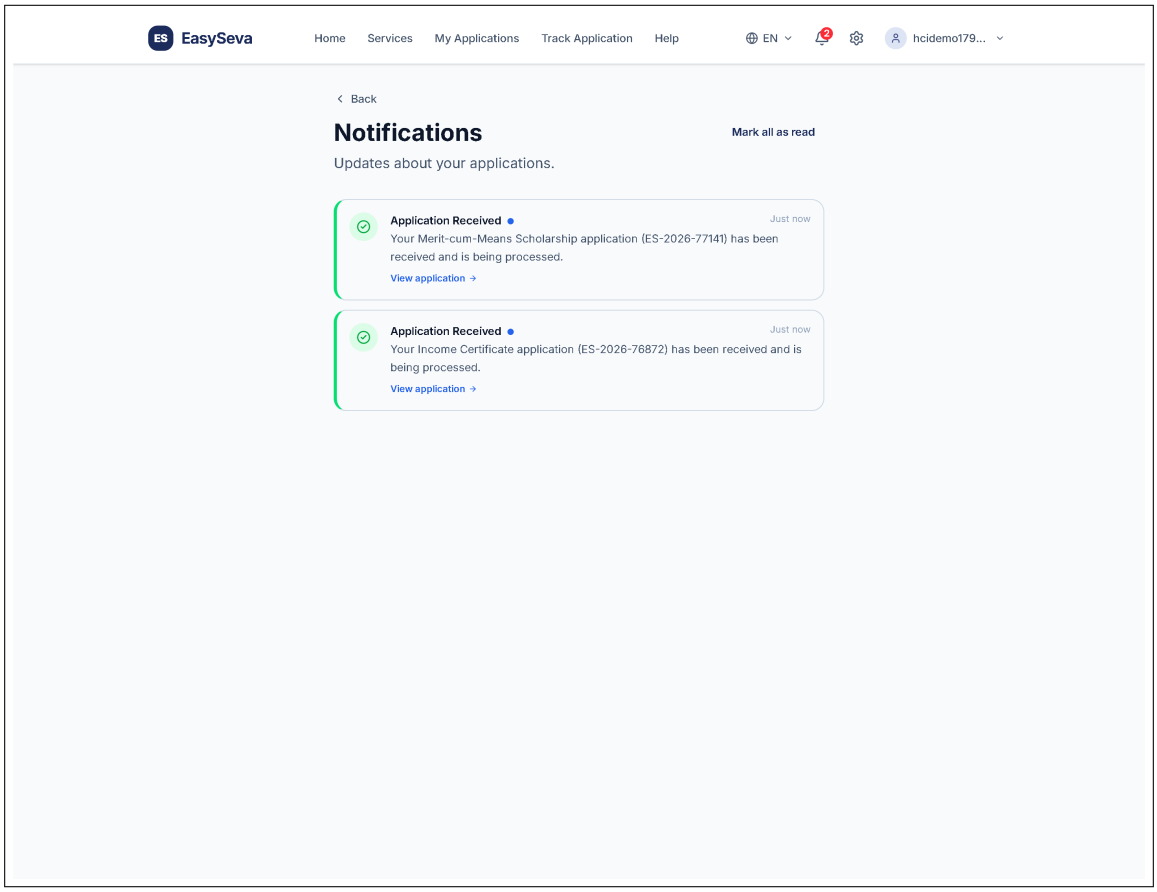


Figure 17. Notifications keep the citizen informed of each application event, with unread indicators and the ability to mark items as read.

Purpose. Surface changes early so the user is never left guessing.

- **Visibility of system status (N1, S3).** *Give proactive feedback.*
  Status changes and required actions are pushed to the user rather than hidden away.
- **User control and freedom (N3, S7).** *Let users manage the feed.*
  Users can mark one item or all items as read, so they stay in control.
- **Recognition rather than recall (N6).** *Signal what is new.*
  An unread count on the header bell and distinct styling per item make new information obvious.

5.13 Profile

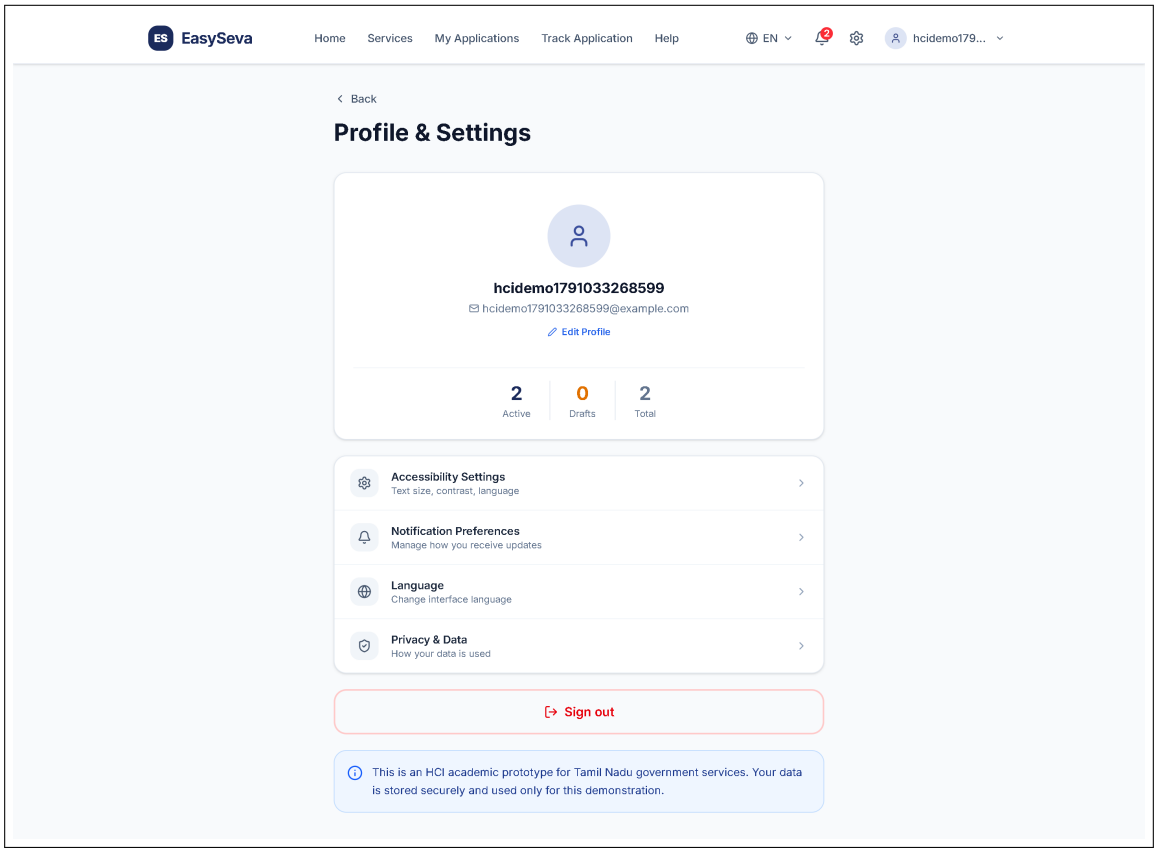


Figure 18. The profile page lets users view and edit their contact details, with clear feedback when changes are saved.

Purpose. Give users ownership of their own account information.

- **User control and freedom (N3, S7).** *Let users own their data.*
  Users edit their name and phone number directly.
- **Informative feedback (S3, N1).** *Confirm the save.*
  Saving shows an explicit changes saved message.
- **Error prevention (N5).** *Validate contact details.*
  Inputs are checked before they are accepted.

5.14 Sign In with a One Time Code

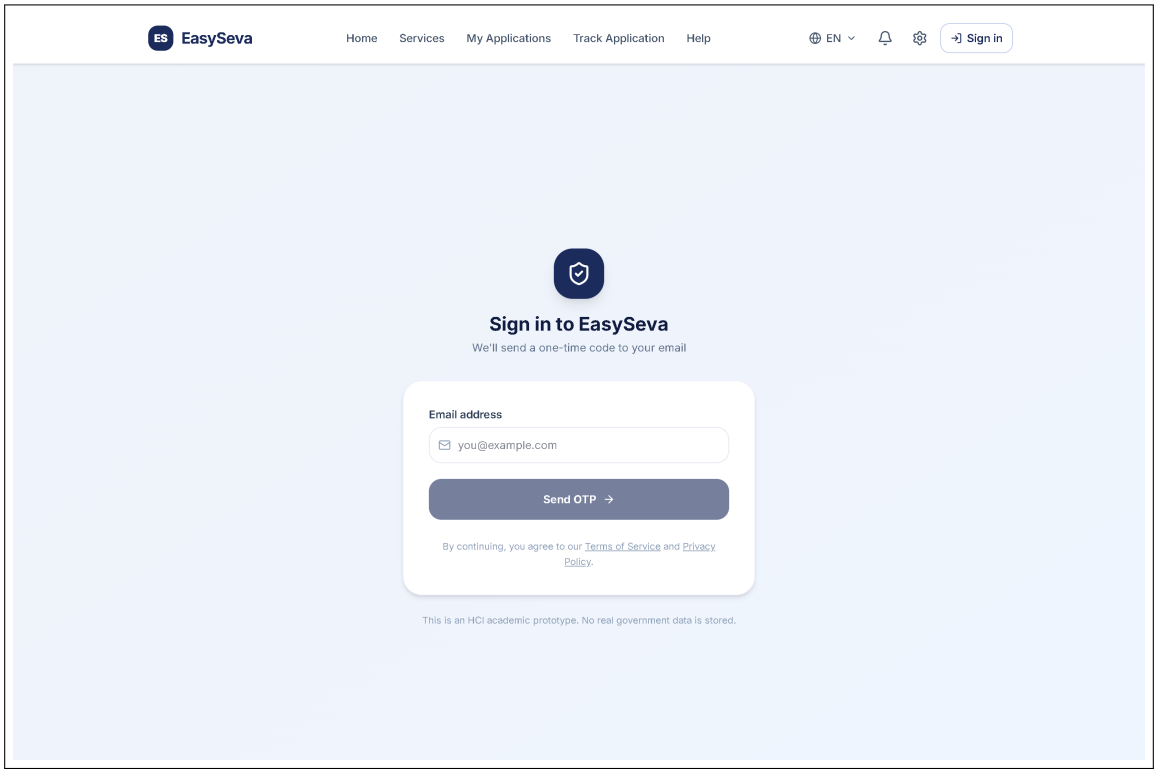


Figure 19. Passwordless sign in. The user enters only an email address.

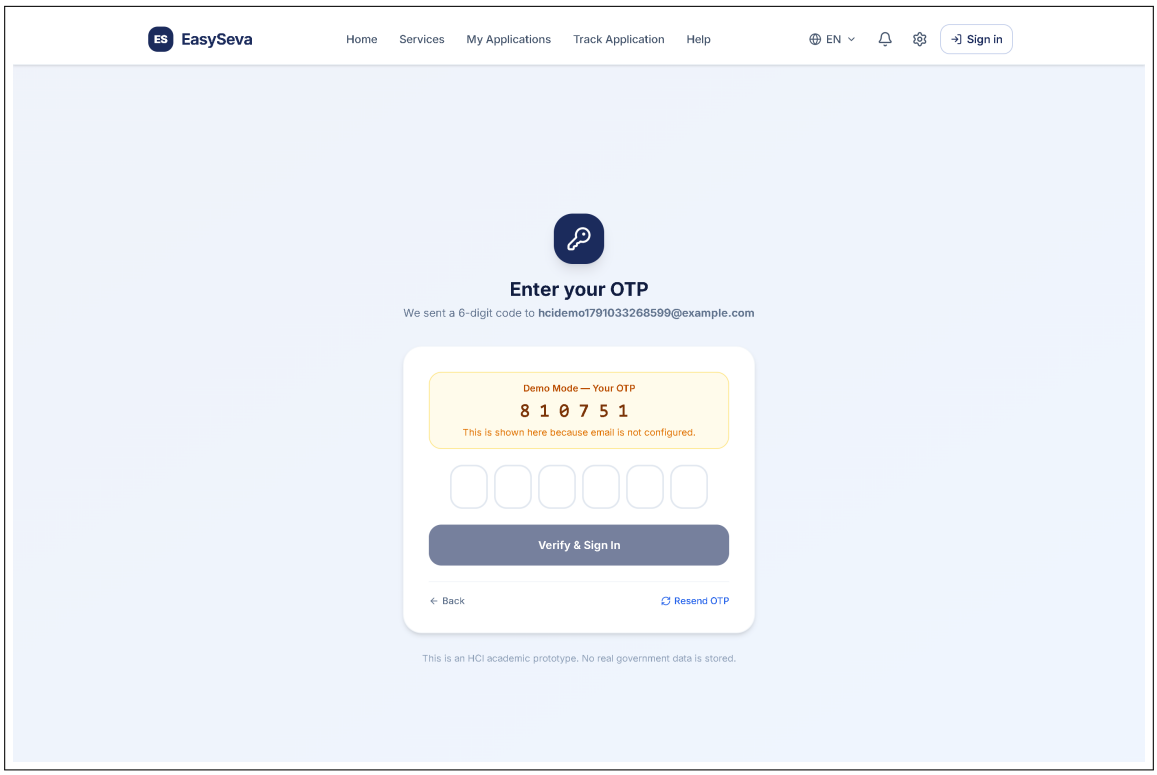


Figure 20. The verification screen with a six box code input, a resend option with a cooldown, and, in demo mode, the code shown on screen.

Purpose. Sign users in with as little friction as possible and with little chance of being locked out.

- **Error prevention (N5, S5).** *Remove a whole failure mode.*
  A one time code means there is no password to forget, reuse, or mistype.
- **Flexibility and efficiency (N7).** *Make entry fast.*
  The six digit code can be pasted and spreads across the boxes on its own, and the focus moves forward automatically.
- **Visibility of system status (N1, S3).** *Show every state.*
  The sending, verifying, and verified states are each shown clearly.
- **Help users recover from errors (N9).** *Recover gracefully.*
  A wrong code clears the boxes with a clear message, and the resend option has a visible cooldown so users are not left stranded.
- **Affordance (D1).** *Make the input obvious.*
  Six separate boxes invite a six digit code and signal exactly how much to type.

5.15 Help Centre

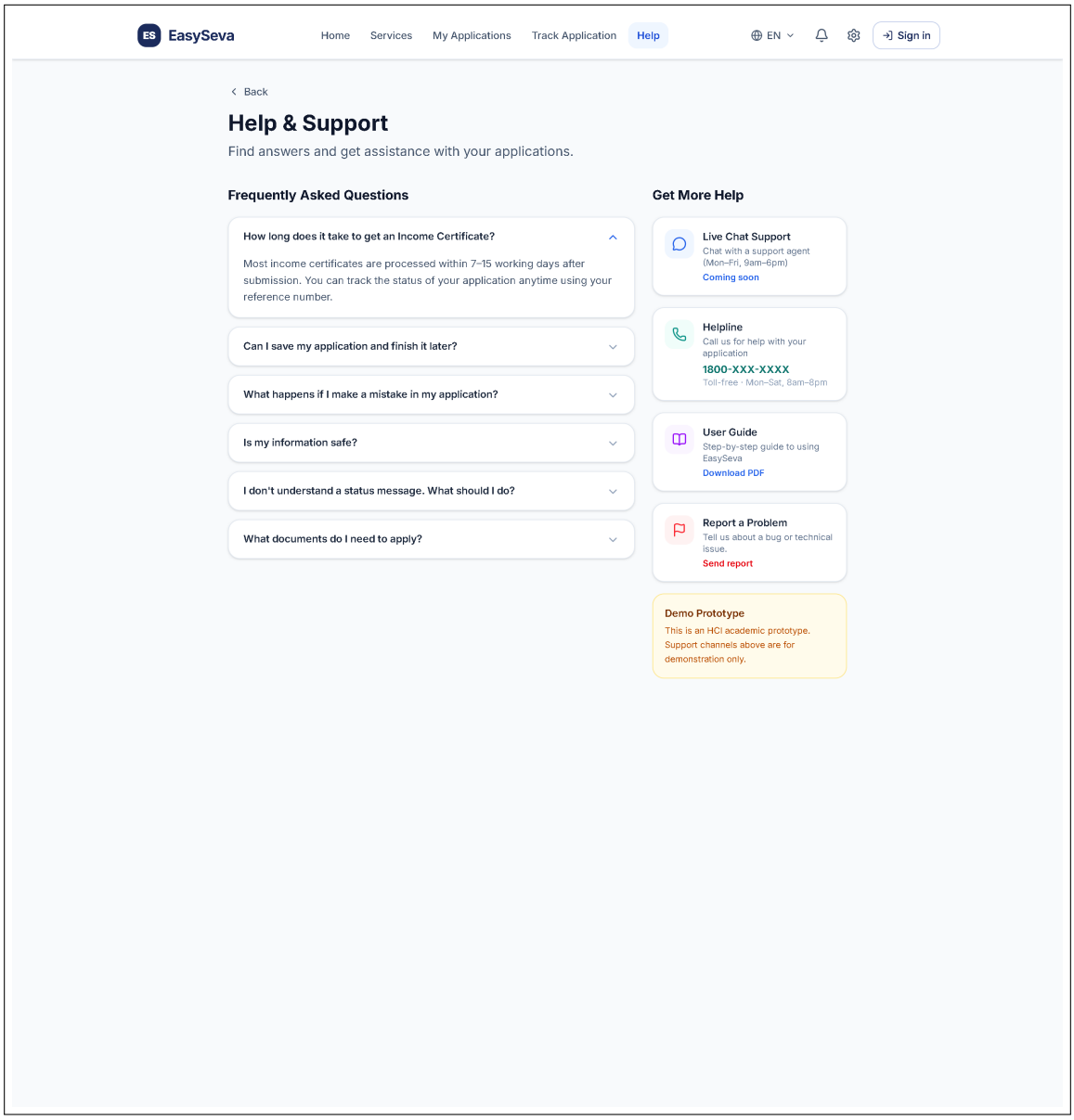


Figure 21. The Help page answers common questions in plain language.

Purpose. Provide support that is easy to search and focused on the task.

- **Help and documentation (N10).** *Make answers findable.*
  Common questions are organised and phrased the way citizens actually ask them.
- **Match with the real world (N2).** *Answer in plain words.*
  The answers avoid jargon and describe actions in the user's terms.

5.16 Responsive and Mobile Experience

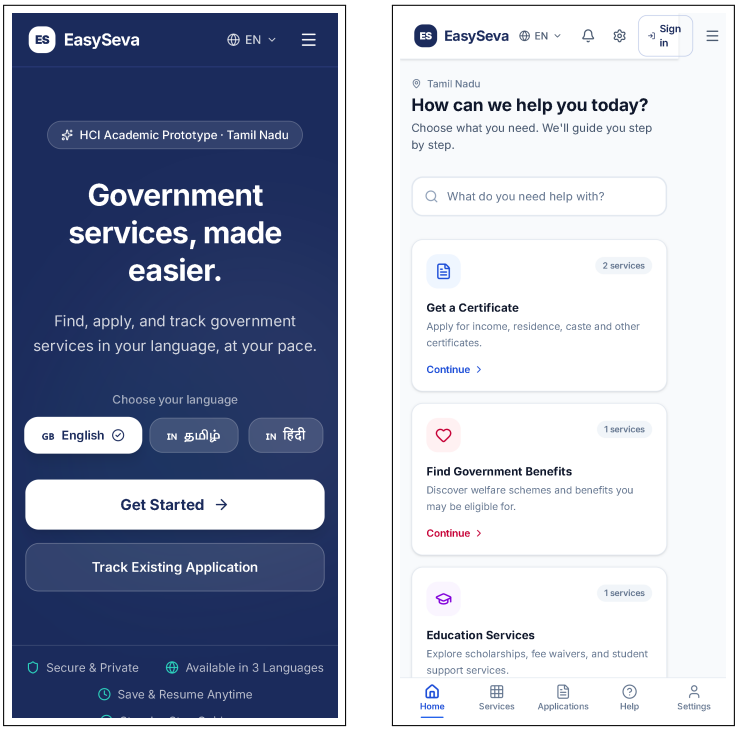


Figure 22. The landing page and the home dashboard on a phone. The layout reflows to a single column and a bottom navigation bar appears so the main actions are within easy reach of the thumb.

Purpose. Serve the majority of citizens who use their phone.

- **Flexibility and efficiency (N7).** *Work on any device.*
  Layouts move from several columns on a desktop to a single column on a phone without losing any function.
- **Cater to universal usability (S2).** *Suit touch and reach.*
  A bottom navigation bar and large touch targets suit one handed use.
- **Consistency and standards (N4, S1).** *Keep the same mental model.*
  The mobile and desktop versions share components and wording, so skills carry over.

6 Cross Cutting Principles

Some principles are not tied to a single screen but run through the whole product.

- **Help users recover from errors (N9).** *Give plain and constructive errors.*
  Error messages everywhere are written in human language and suggest a next step, for example please enter a valid ten digit mobile number, or a code error that clears the field and invites another try. Network problems fall back gracefully so the user is never stuck.
- **Keep users in control (S7, N3).** *Stay predictable.*
  Actions are started by the user. Anything that cannot be undone, such as a final submit, needs an explicit confirmation, and can be approached through a reversible draft first.

- **Consistency (N4, S1, D5).** *Use one design language.*
  Shared components and tokens make every button, card, badge, and field behave the same way across the app.
- **Accessibility and POUR.** *Be inclusive by default.*
  Keyboard operation with visible focus, a skip link, ARIA roles and labels, scalable text, a high contrast theme, and reduced motion support make the app usable by a far wider range of people.
- **Conceptual model (D6).** *Give a clear mental picture.*
  The same five stage journey of discover, check, prepare, fill, and confirm gives users an accurate idea of how applying works, and it carries across all nine services.

7 Heuristic Coverage Table

Table 1 maps each Nielsen heuristic to places where it is applied in EasySeva.

Table 1. Where each usability heuristic is applied.

Heuristic	Where it is used in EasySeva
Visibility of system status	Progress bars, step indicators, status badges and the timeline, loading, saving, and verifying states, and a confirmation with a reference number.
Match with the real world	Task based categories, plain language labels and statuses, worked examples, and rupee formatting.
User control and freedom	Back links and breadcrumbs, save and finish later, an editable review, a cancellable confirmation dialog, and continue anyway on eligibility.
Consistency and standards	Shared Button, Card, Badge, and Input pieces and design tokens across every page.
Error prevention	An eligibility pre check, inline field checking, document type and size checks, a confirmation before submit, and a one time code instead of a password.
Recognition rather than recall	Document examples, the review summary, breadcrumbs, visible service cards, and unread cues.
Flexibility and efficiency	Three languages, search and a guided finder, code paste, tracking without login, and keyboard navigation.
Aesthetic and minimalist design	One primary action per screen, a chunked catalogue and form, and an uncluttered first screen.
Recover from errors	Human readable messages with a next step, a clear and retry for the code, and graceful network fallbacks.
Help and documentation	Service questions and answers, the Help centre, and the why are we asking and what can I use notes.

8 Conclusion

EasySeva shows that the difference between an intimidating government portal and one that actually helps people comes down, for the most part, to applied Human Computer Interaction. By grounding every screen in ideas that are well established, such as keeping status visible, speaking the user's language, preventing errors before they happen, favouring recognition over memory, keeping the user in control, and designing for everyone, the application turns a rare and stressful task into one that feels calm, guided, and forgiving.

These principles are not decoration. Each one maps to a real reduction in the failures that citizens face, including abandoned forms, rejected documents, lost progress, and confusion about status. Good design, in other words, is what lets technology truly serve people. EasySeva is a working and deployed example of that idea.